MIZORAM PUBLIC SERVICE COMMISSION

COMPETITIVE EXAMINATION FOR RECRUITMENT TO THE POST OF ASSISTANT DIRECTOR (FOOD & DRUGS) UNDER HEALTH & FAMILY WELFARE DEPARTMENT, GOVERNMENT OF MIZORAM. (NOVEMBER, 2025)

TECHNICAL PAPER-II

Time Allowed : 3 hours FM : 200

SECTION - A *(Conventional Type) (100 Marks)*

This Section should be answered only on the Answer Sheet provided.
Marks for each question is indicated against it.

1. What is the primary objective of Narcotic Drugs and Psychotropic Substances (NDPS) Act? Explain the procedures for cultivation, manufacture, possession, and transportation of narcotic drugs under the NDPS Act. **(2+8=10)**

2. Explain the objectives and classification of medical devices under the Medical Devices Rules, 2017. Discuss the categories of diseases and conditions for which advertisements are prohibited under Drugs and Magic Remedies (Objectionable Advertisements) Act. **(5+5=10)**

3. Explain the principles of drug metabolism. Describe Phase I and Phase II reactions with examples of each category. **(4+6=10)**

4. Explain classical and non-classical bioisosterism with appropriate examples. Discuss how bioisosterism is applied in drug design to improve pharmacodynamic and pharmacokinetic properties of drugs. **(5+5=10)**

5. Differentiate between rigid and flexible docking. Discuss the concept of pharmacophore mapping and pharmacophore based screening of small molecules. **(2+8=10)**

6. Explain the need and importance of quality control in drugs of natural origin. Describe the chemical classes, sources, and therapeutic uses of flavonoids with examples. **(4+6=10)**

7. Discuss the role of WHO guidelines in maintaining the quality and purity of herbal drugs. Explain the ICH guidelines for stability testing of herbal formulations. **(5+5=10)**

8. Write about the salient features of Drugs and Magic Remedies Act and its Rules. Describe the general requirements of labeling under Drugs and Cosmetics Act, 1940. **(10)**

9. Define nutraceutical. Explain the health benefits and role of nutraceutical in cardiovascular diseases. Write the possible side effects and interactions of garlic. **(10)**

10. Define and differentiate between adulterated, misbranded and spurious drugs as per the D&C Act. Explain the correponding penalties for these offences. **(10)**

SECTION - B *(Multiple Choice questions) (100 Marks)*

All questions carry equal mark of 2 each. Attempt all questions.
*This Section should be answered only on the **OMR Response Sheet** provided.*

1. Schedule X drugs are:
(a) Cosmetics
(b) Homeopathic drugs
(c) Narcotic and psychotropic drugs
(d) Ayurvedic drugs

2. Which of the following conditions does not make a drug misbranded?
(a) False or misleading claims on the label
(b) Concealment of damage through coating
(c) Incorrect strength or composition
(d) Improper labelling

3. The formula for calculating the ceiling price (Pc) is:
(a) $Pc = Ps \times (1 + M/100)$
(b) $Pc = Ps \times (1 - M/100)$
(c) $Pc = Ps + M$
(d) $Pc = M \times 100 / Ps$

4. Under the Schedule of Medicinal & Toilet Preparations (Excise Duties) Act, 1955**,** toilet preparations containing alcohol or narcotic drug attract a rate of duty—
(a) 20% *ad valorem*
(b) 4% *ad valorem*
(c) 50% *ad valorem*
(d) 10% *ad valorem*

5. The Drugs (Prices Control) Order, 2013 was issued under which Act?
(a) Drugs and Cosmetics Act, 1940
(b) Essential Commodities Act, 1955
(c) Pharmacy Act, 1948
(d) Drug Price Regulation Act, 1970

6. Hydrogen bonding in a drug molecule primarily influences:
(a) Drug metabolism through Phase II reactions
(b) Partition coefficient in non-polar solvents
(c) Binding affinity to biological targets such as enzymes and receptors
(d) The rate of ionization in aqueous solution

7. In QSAR, the Hammett constant (σ) describes:
(a) Lipophilicity
(b) Steric hindrance
(c) Electronic effects of substituents
(d) Hydrogen bonding capacity

8. A pharmacophore model includes:
(a) Only the structure of the enzyme
(b) The essential features responsible for biological activity
(c) The full 3D structure of the drug molecule
(d) Only hydrogen bonding acceptor groups

9. In molecular docking, the scoring function is used to:
(a) Identify the enzyme classification
(b) Rank the protein structures
(c) Estimate the binding affinity between ligand and target
(d) Modify the drug's solubility

10. Which of the following changes would most likely reduce protein binding of a drug?
(a) Increasing lipophilicity
(b) Adding a hydrophilic functional group
(c) Enhancing aromaticity
(d) Introducing a halogen substituent

11. The Taft steric parameter (Es) quantifies:

(a) Lipophilic contribution to activity
(b) Electronic influence of substituents
(c) Spatial or bulk effects of substituents
(d) Hydrogen bonding capability

12. The pharmacological activity of enantiomers can differ because:

(a) Only one enantiomer binds to plasma proteins
(b) Enzymes and receptors are chiral
(c) They have different chemical formulas
(d) They are always metabolized at the same rate

13. The primary goal of Phase II metabolism is to:

(a) Decrease the molecular weight of the drug
(b) Inactivate the drug completely
(c) Convert drugs into more lipophilic forms
(d) Increase water solubility for excretion

14. *Papaver somniferum* yields alkaloids belonging to which class?

(a) Indole alkaloids
(b) Pyrrolidine alkaloids
(c) Quinoline alkaloids
(d) Benzylisoquinoline alkaloids

15. Lycopene, a carotenoid found in tomatoes, is beneficial in—

(a) Alzheimer's disease
(b) Prostate cancer prevention
(c) Diabetes
(d) Asthma

16. *Hypericum perforatum* (St. John's Wort) induces—

(a) CYP3A4 enzymes
(b) CYP2C19 inhibition
(c) MAO inhibition
(d) CYP2D6 inhibition

17. *Ephedra* use with caffeine may cause—

(a) CNS depression
(b) Hypotension
(c) Drowsiness
(d) Hypertensive crisis

18. The Isoprene Rule states that:

(a) Terpenes are made up of linked glucose units
(b) Terpenes are composed of C5 units arranged head-to-tail
(c) Alkaloids are derived from amino acids
(d) Flavonoids are formed by C6-C3-C6 unit

19. Which of the following is *not* WHO-recommended method for evaluation of herbal drugs?

(a) Macroscopic and microscopic evaluation
(b) Chromatographic fingerprinting
(c) Radiographic imaging
(d) Assay of marker compounds

20. ICH guidelines recommend testing herbal drug stability under accelerated conditions for:

(a) 6 months
(b) 1 month
(c) 1 year
(d) 2 months

21. Which nutraceutical is most commonly used for lowering cholesterol and improving heart health?

(a) Omega-3 fatty acids
(b) Beta-carotene
(c) Probiotics
(d) Coenzyme Q10

22. Iridoids are classified chemically as:

(a) Monoterpenoids
(b) Sesquiterpenoids
(c) Diterpenoids
(d) Triterpenoids

23. The primary mechanism of action of cardiac glycosides is:

(a) Activation of beta-adrenergic receptors
(b) Calcium channel blockade
(c) Angiotensin II inhibition
(d) Inhibition of Na^+/K^+ -ATPase pump

24. Shade drying is preferred over sun drying because it:

(a) Preserves color and active constituents
(b) Reduces moisture more rapidly
(c) Kills microbes effectively
(d) Increases aroma

25. Drugs covered under this schedule are not permitted for repacking license. Identify the correct schedule.

(a) Schedule H
(b) Schedule G
(c) Schedule O
(d) Schedule C and C1

26. Chose the CORRECT statement with respect to *'The Pharmacy Act, 1948'*:

(a) Education regulation 1991 does not prescribe the minimum qualification for the registration as Pharmacist
(b) Section 12 of the Act deals with the approval of course of study under Chapter 2 there of
(c) Section 12 of the Act deals with the approval of course of study and examination under Chapter 2 there of
(d) State government is authorized to make any rules with respect to course of study

27. Penalty for the cultivation of any cannabis plant to produce, sell, purchase transport in contravention of Narcotic Drugs and Psychotropic Substances Act and Rules on first conviction is:

(a) Rigorous imprisonment up to 10 years or fine up to Rs. 10 Lakhs
(b) Rigorous imprisonment up to 10 years or fine up to Rs. 1 Lakh
(c) Rigorous imprisonment up to 6 months
(d) Fine up to Rs. 10 Lakhs

28. What is the effect of ionization on drug absorption?

(a) Increases absorption
(b) Decreases absorption
(c) No effect on absorption
(d) Depends on the pH

29. Which of the following proteins is primarily involved in drug binding?

(a) Albumin
(b) Globulin
(c) Fibrinogen
(d) Hemoglobin

30. What is the effect of a fat-free diet on cytochrome P450 levels?

(a) Increases cytochrome P450 levels
(b) Variable effect on cytochrome P450 levels
(c) No effect on cytochrome P450 levels
(d) Decreases cytochrome P450 levels

31. The most common pharmacological effect of tobacco smoking on drug metabolism is:

(a) Enzyme inhibition
(b) Enzyme induction
(c) Enzyme saturation
(d) No effect

32. The most favorable bonding between ligand and protein is:

(a) Covalent bonding
(b) Metal Binding
(c) Vander Waals Bonding
(d) Hydrogen bonding

33. What is the purpose of the Hansch equation?

(a) To predict the color of a compound
(b) To predict the biological activity of a compound based on its structure
(c) To predict the solubility of a compound in water
(d) To predict the melting point of a compound

34. Which of the following is NOT a hydrophobic parameter?

(a) Partition coefficient (P)
(b) Molecular connectivity
(c) Pi Substituent constant
(d) Log P

35. Hypericum is known to interact with certain medications because it can:

(a) Increase the activity of liver enzymes
(b) Decrease the activity of liver enzymes
(c) Have no effect on liver enzymes
(d) Only interact with herbal supplements

36. Ginkgo biloba is used for its:

(a) Expectorant activity
(b) Lipid lowering activity
(c) PAF antagonistic activity
(d) Antidepressant activity

37. What is polyploidy?

(a) A condition where an organism has two sets of chromosomes
(b) A condition where an organism has more than two sets of chromosomes
(c) A condition where an organism has one set of chromosomes
(d) A condition where an organism has no chromosomes

38. The drug used for irritable bowel syndrome is:

(a) Ispaghula
(b) Strophanthus
(c) Nux vomica
(d) Ephedra

39. In the Drugs and Cosmetic Act and Rules, the Schedule relating to GMP is:

(a) Schedule M
(b) Schedule C
(c) Schedule Y
(d) Schedule H

40. Two ex-officio members of the Drugs Technical Advisory Board under Drugs and Cosmetics Act are:

(P) The Drugs Controller General of India
(Q) The President, Medical Council of India
(R) The Secretary, Pharmacy Council of India
(S) The Director, National Institute of Pharmaceutical Education and Research, India

(a) P, Q
(b) P, R
(c) R, S
(d) P, S

41. The Drug Price Control Order (DPCO) is an order issued by the Government under the _____ which enables it to fix the prices of some essential bulk and their formulations

(a) Essential Commodities Act
(b) Essential Commodities Amendment
(c) Essential Commodities Accessories
(d) Ethical Commodities Act

42. As per the Medical Termination of Pregnancy Act and Rules, the safe custody of 'Forms' is with:

(a) Standing committee
(b) Registered Medical Practitioner
(c) Owner of the approved place
(d) Chief Medical Officer

43. Who has the power to fix the ceiling price of scheduled formulations?

(a) National Pharmaceutical Pricing Authority
(b) Director General Health Services
(c) National Medical Commission
(d) Pharmacy Council of India

44. Medical devices are classified into how many classes under the Rules?

(a) Two (b) Three

(c) Four (d) Five

45. As per Medical Termination of Pregnancy Act, 1971; the term abortion indicates:

(a) Termination of pregnancy after foetus is viable

(b) Termination of pregnancy of less than 12 weeks

(c) Termination of pregnancy before foetus is viable

(d) Termination of pregnancy of more than 12 weeks

46. The objective of Drug and Remedies Act is:

(a) To promote certain types of advertisements related to drugs which mislead public

(b) To prohibit advertisements related to magic remedies with false claims

(c) To provide uniform education and training to pharmacists

(d) To regulate profession of pharmacy

47. Which section of the Pharmacy Act, 1948 empowers the PCI to prescribe minimum standards of education required for qualification as a pharmacist?

(a) Section 10 (b) Section 11

(c) Section 12 (d) Section 16

48. The term 'psychotropic substance' is defined in section:

(a) 2(xxiii) (b) 2(ix)

(c) 2(xi) (d) 2(viii)

49. Diseases prohibited from advertisement under Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954 are listed in:

(a) Schedule A (b) Schedule J

(c) Schedule H (d) Schedule K

50. The application for import licence for a medical device is made in —

(a) Form MD-10 (b) Form MD-14

(c) Form MD-25 (d) Form MD-26

* * * * * * *